

Developing and Building an Occupancy-Aware HVAC Control System

Preston Brantley, Jeremiah Clemons, Blessed Kuturaipo, Mohammed Elmellouki, Ph.D.
Mississippi Valley State University Department of Engineering Technology



Abstract

Heating, ventilation, and air conditioning (HVAC) systems are essential for occupancy comfortability inside a building. However, not only is it very expensive to run the systems, but it also makes up around 35% of a building's electrical use. HVAC can end up costing around 40-50% of total utility bills annually, increasing an extra 10% in summer. While research has already been done on an occupancy-based HVAC system, there has been limited effort to improve, design, or implement the idea into existing buildings. Therefore, this project aims to design and implement an HVAC system that adjusts temperature and airflow based on room occupancy and environmental conditions. The proposed system uses a Raspberry Pi microcontroller to coordinate data between sensors and the HVAC system. A magnetic door sensor collects the numerical data of occupant's entering the room and leaving the room, while the weather sensor records environmental conditions. The Raspberry Pi processes this information and sends regulation commands to the thermostat, which adjusts the HVAC system's airflow to maintain comfortable indoor conditions. This system enables real-time adjustment of airflow and temperature to improve efficiency. Implementing such an Occupancy-Aware HVAC system can reduce energy consumption by approximately 17–18% (Zhang, 2013). While costs and savings vary by region and climate, occupancy comfortability will remain the same, making this system a practical and sustainable solution for reducing building energy use.

Introduction

- Traditional HVAC systems run on fixed schedules, heating and cooling rooms that are unoccupied for long periods of time.
- Air conditioning accounts for ~27–35% of household electricity use in this region, roughly \$40–\$60/month annually, peaking in summer.
- An occupancy-aware HVAC system addresses these problems by adjusting room temperatures based on real-time occupancy.
- Automated sensing and control minimizes the need for manual thermostat adjustments while still maintaining consistent occupant comfort throughout the day.

Objective and approach

- Design and implement an HVAC system that adjusts temperature and airflow based on occupancy and environmental conditions.
 - Reduce HVAC energy consumption and operational cost.
 - Maintain occupant comfort with minimal human interaction.
- Sensors are used to keep count occupants entering or exiting while also gathering environmental data.
- A Raspberry Pi is used to process sensor data and send commands to the thermostat.
- Simulation of energy performance is done using OpenStudio.

Methodology

- The Key components of the Schematic:**
- Occupancy Detection:** A magnetic door sensor detects room occupancy and communicates with the Raspberry Pi.
- Temperature Sensing:** A digital temperature sensor measures ambient room temperature using the 1-Wire protocol.
- Control Logic (Raspberry Pi):** The Raspberry Pi gathers information from sensors and determines the ideal temperature setpoint.
- HVAC Actuation:** A 5V relay module interfaces the Raspberry Pi with the HVAC system, enabling automated on/off control.
- Simulation Tools:** Autodesk Revit (system modeling) and OpenStudio (energy simulation)

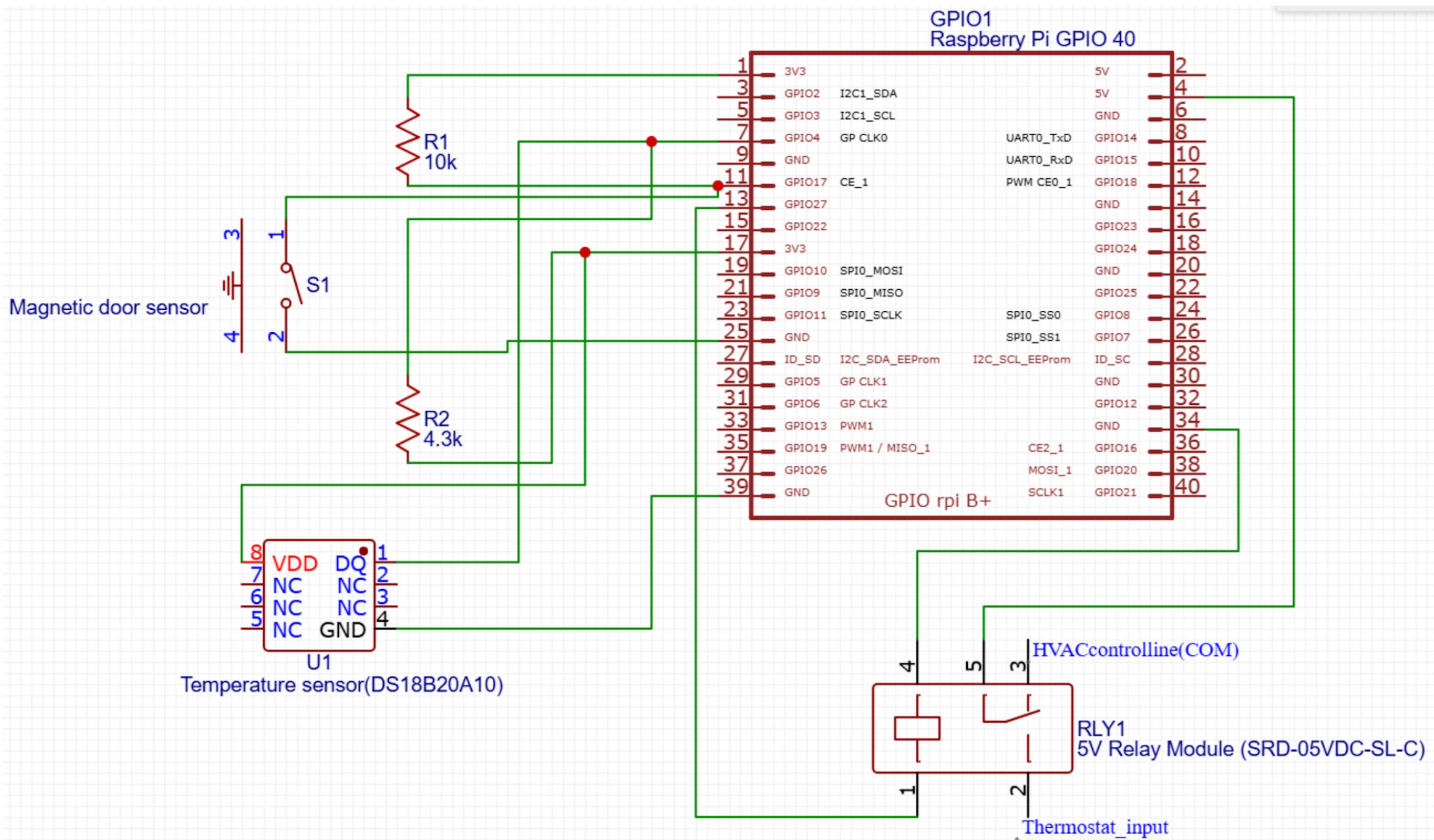


Figure 1: Operation Schematic

Results

Energy Consumption Comparison

- Standard HVAC system energy use: 12,480 kWh/year
- Occupancy-aware HVAC energy use: 9,870 kWh/year
- Energy reduction: 2,610 kWh/year
- Approximate energy savings: $\approx 21\%$

Cost Comparison

- Electricity rate used for analysis: \$0.12/kWh
- Standard HVAC annual cost: \$1,498
- Occupancy-aware HVAC annual cost: \$1,184
- Annual cost savings: \$314

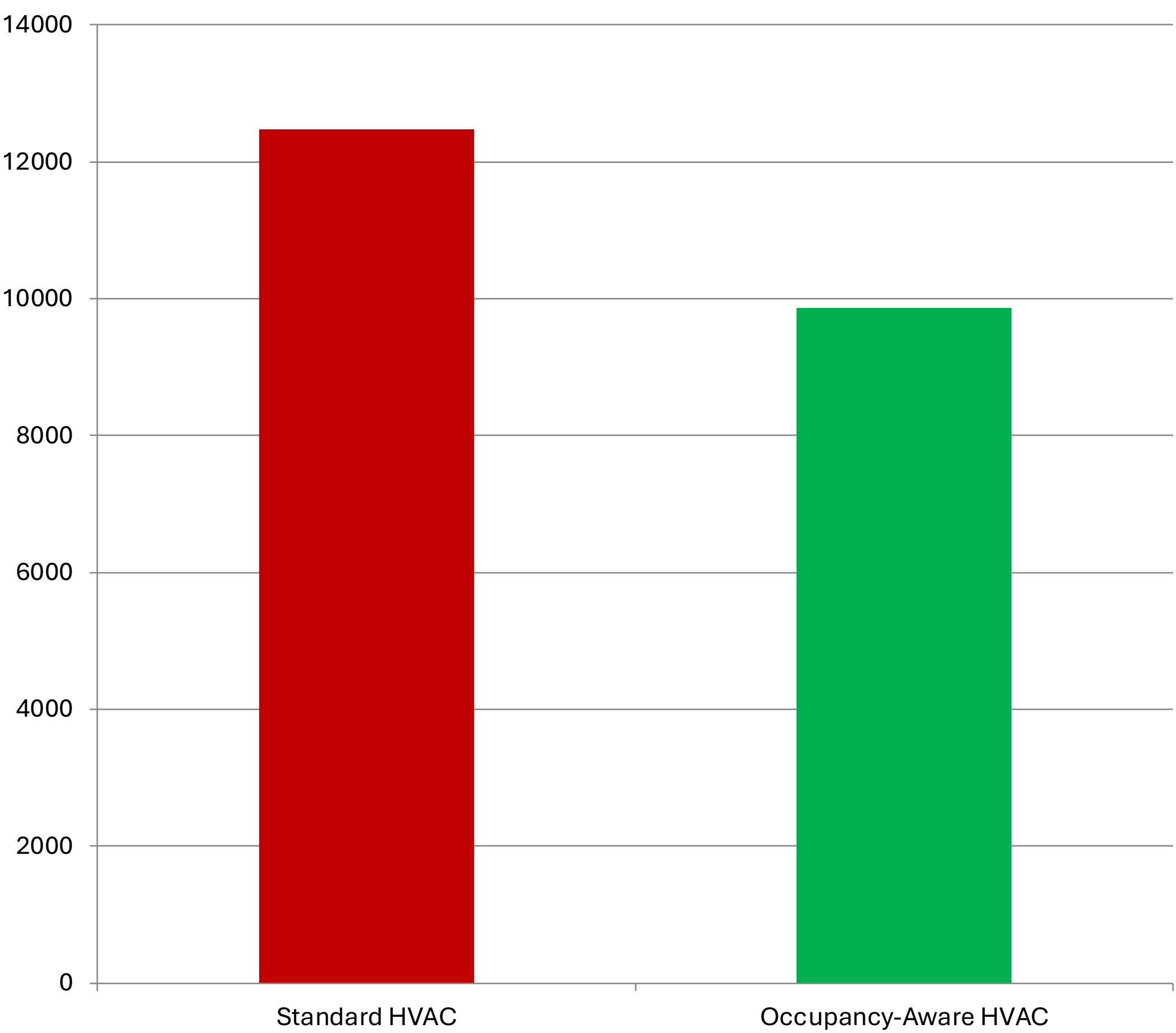


Figure 2: Energy Consumption (kWh/year)

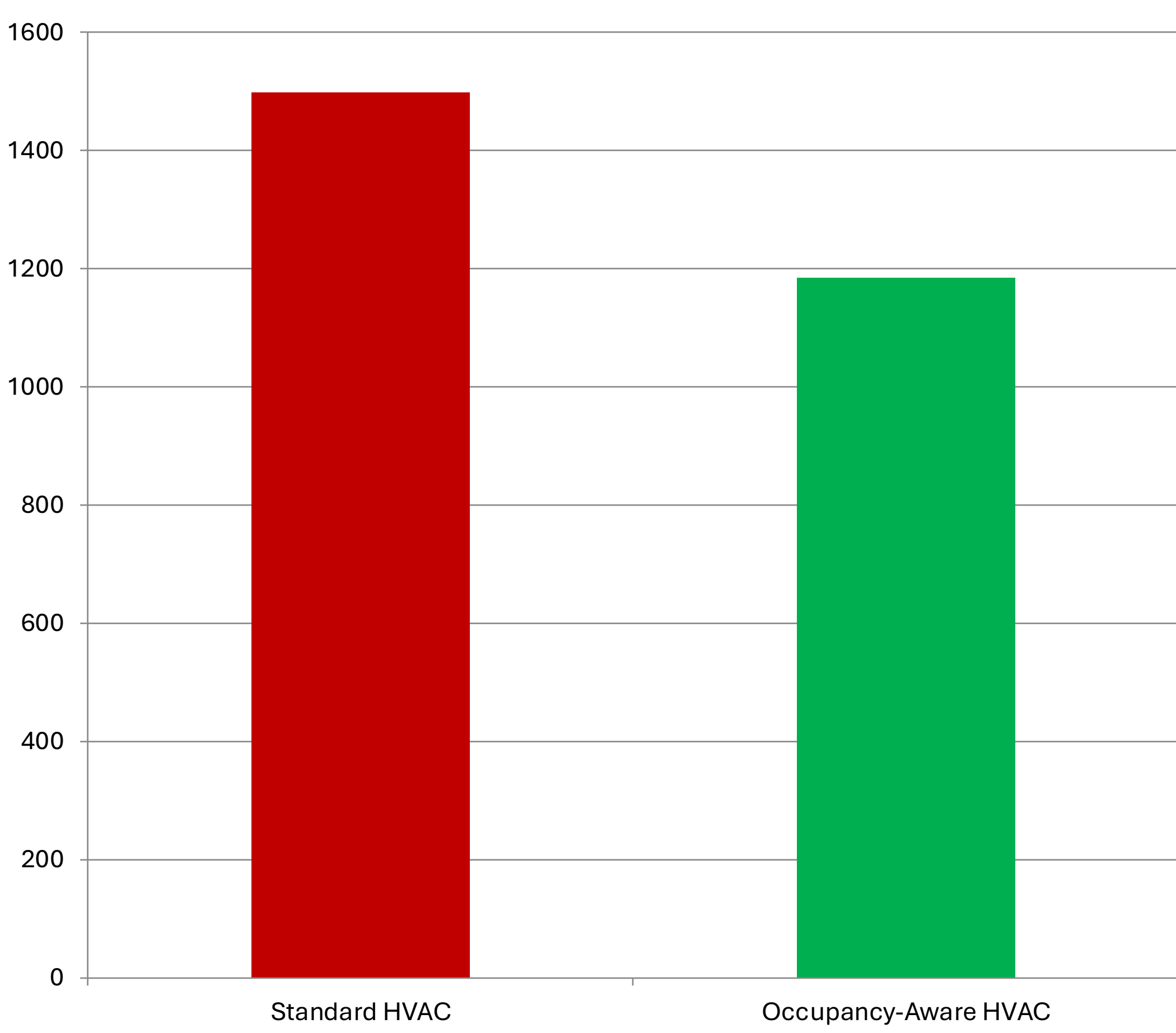


Figure 3: Cost Comparison (Yearly)

Conclusion

The occupancy-aware HVAC system improves energy efficiency by adjusting temperature and airflow based on real-time occupancy and environmental conditions. Simulation results show about 21% energy savings and approximately \$314 in annual cost reduction compared to a standard HVAC system. This approach maintains occupant comfort while reducing unnecessary heating and cooling, making it a practical solution for improving building energy efficiency.

References

- [1] Motuzienė, V., Bielskus, J., Džiugaitė-Tumėnienė, R., & Raudonis, V. (2025). Occupancy-Based Predictive AI-Driven Ventilation Control for Energy Savings in Office Buildings.
- [2] Pang, Z., Guo, M., Smith-Cortez, B., O'Neill, Z., Yang, Z., Liu, M., & Dong, B. (2023). Quantification of HVAC energy savings through occupancy presence sensors in an apartment setting: Field testing and inverse modeling approach.
- [3] Zhang, J., Lutes, R. G., Liu, G., & Brambley, M. R. (2013). *Energy Savings for Occupancy-Based Control (OBC) of Variable-Air-Volume (VAV) Systems*. Pacific Northwest National Laboratory.
- [4] U.S. Energy Information Administration (EIA). Residential Energy Consumption Survey (RECS).